Technical Description

Test Project - PH Desktop

09 IT Software Solution for Business

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Introduction

In this session, you will take on the role of a full-stack developer, responsible for building the front-end user interface (UI) and the back-end for Vendor Management software system Category with Philippines' Food Festival theme.
The Vendor Management Desktop Application will be designed to streamline the procurement and management of vendors participating in the Philippines Food Festival. The application will allow event organizers to manage vendor information and generate vendor reports.
This session is designed to assess your skills in:

Front-End Development: Designing and implementing user interfaces that are user-friendly, and visually appealing.

Back-End Development: Develop API that adheres to industry best practices, handles data interactions securely, and provides a seamless experience for the front-end application.

Integration: Ensuring smooth communication and data exchange between the front-end and back-end components.

Problem-Solving: Identifying and resolving technical challenges that arise during development.

Contents

This session package includes the following materials:

Session Instructions (PDF): Detailed instructions outlining the tasks to be completed and deliverables expected for this session.

Common Folder: This folder contains additional resources such as logo, icons, style guide, and other design assets that can be used throughout the development of the application.

Database Schema (SQL): A SQL script containing the structure for the tables which you will use in this session.

Instructions to the Competitor

3.1 Database Setup and Data Import

Objective:

The objective of this project is to develop a vendor management system that allows users to:

Handle successful cleaning and importing of provided database.

Manage vendors who provide Filipino food desserts and delicacies for a food festival.

Handle vendor information, product details, and orders.

Provide a user-friendly interface for data entry and retrieval.

Tasks:

1. **Database Creation:** Create a database named FoodFestivalDB (or a similar name).
2. **Schema Execution:** Execute the provided SQL script.
3. **Data Import:** Import the provided .csv files.

Deliverables:

A text file (PHVMngt_DatabaseCredentials.txt) containing the connection string or credentials required to access your database (server, database, username, password).

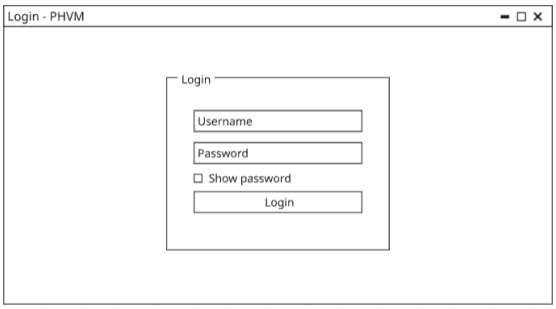
Integration of the database into the application.

3.2 Authentication and Access Control

Objective:

Implement secure user authentication and role-based access control system for the Philippine Food Festival Vendor Management Application, ensuring data protection and efficient user administration.

Tasks:



1. **Authentication**
   - Login form with username and password fields with show/hide toggle button for password.
   - Error handling for empty fields, with clear error messages.
   - Login form fields must be cleared upon logout.
   - After three consecutive failed login attempts, the user will be temporarily locked out for 10 seconds. There should be a clear indicator and timer.
2. **Role-based Access Control**
   - The "User" level should only be able to edit and delete vendors, products, and orders belonging to them.
   - The "Admin" level can edit and delete all vendors, products, and orders.

Deliverables:

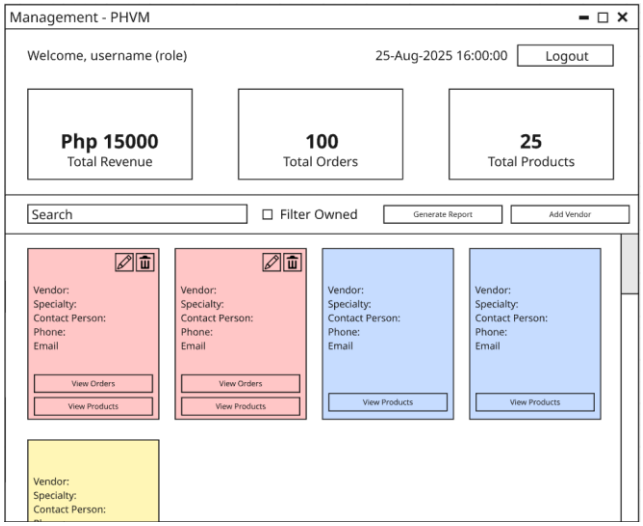
- Login form with authentication and role-based access control.

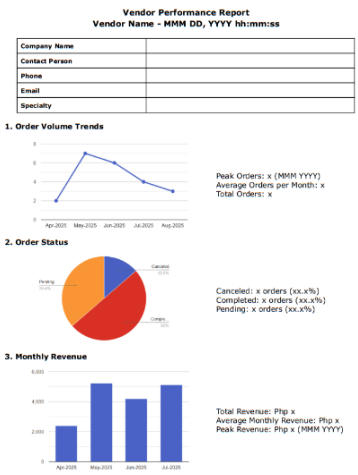
3.3 UI Development – Vendor, Product, and Order Management

Objective:

Develop a user-friendly desktop interface for managing vendors, products, and orders.

Tasks:





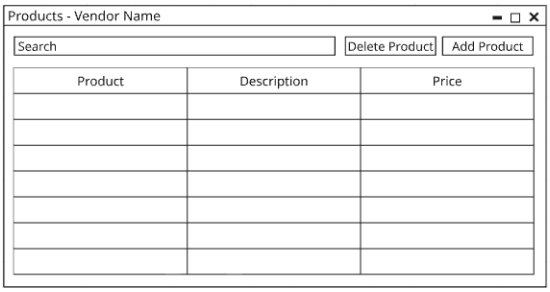
1. Vendor Management Form: This is the main form the client can view all vendors registered to the system, edit owned vendors, and generate reports.

- Welcome Message: Displays the logged-in user's username and role, shown as (Admin) or (User) next to the name.

- Logout Button: Returns the user to the login screen.
- Date & Time: Display the current running date and time in this format (dd-MMM-yyyy HH:mm:ss).
- When a vendor is selected, the following statistics should be updated and displayed: Total Revenue, Total Orders (completed), Total Products.
- A scrollable grid view will display each vendor with the following requirements:
    - The vendor items should be color coded such that each vendor is uniquely identifiable. Use the following colors: (Red, Blue, Yellow)
    - Vendor Details: Display Vendor Name, Specialty, Contact Person, Phone, Email.
    - Delete button: delete the currently selected vendor after a confirmation box. This button is only visible if the vendor belongs to the user, or if the user is an admin.
    - Edit button: open the add/edit vendor form for editing that existing vendor. This button is only visible if the vendor belongs to the user, or if the user is an admin.
    - View Products: Open the Product Management Tab for that vendor.
    - View Orders: Open the Order Management Tab for that vendor. This button is only visible if the vendor belongs to the user, or if the user is an admin.
- The "add vendor" button will open an empty add/edit vendor form for creating a new vendor.
- The search box will filter the list by using either the vendor, specialty, or contact person.
- The "filter owned" checkbox will filter the list into only showing vendors belonging to the user.
- The "generate report" button will generate a report in PDF Format with the following requirements:
    - Overview of Vendor Profile including: Company Name, Contact Person, Phone, Email, Specialty.
    - Line Graph for visualizing Order Volume Trend. Include the following metrics: Peak Orders + Month, Average Orders per Month, Total Orders.
    - Pie Chart for visualizing Order Status. Include the following metrics: Canceled, Completed, and Pending order count with percentages.
    - Bar Graph for visualizing Monthly Revenue. Include the following metrics: Total Revenue, Average Monthly Revenue, Peak Revenue + Month.
    - This button is only clickable if the vendor belongs to the user, or if the user is an admin.

2. Add/Edit Vendor Form: This is the form where the client is able to create new vendors or edit existing vendors.

- Include the following editable fields: Name, Contact Person, Phone Number, Email, Specialty.
- "Save" button for saving changes. These changes must be reflected in the table and the database.
- All fields must be validated when attempting to save. No field should be empty. Phone number should follow a valid landline or mobile format (02XXXXXXXX, 09XXXXXXXXX). Email should be a valid email (example@domain).
- "Cancel" button will discard the changes made.



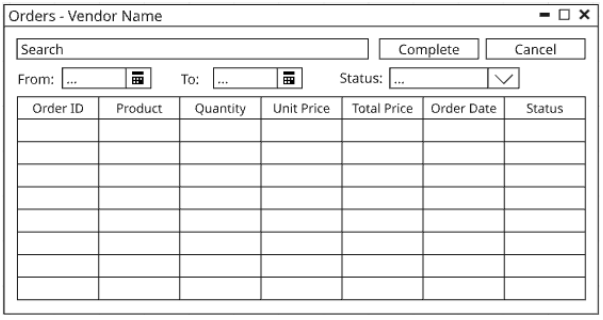
3. Product Management Tab: This is the tab where the client can view all products registered in the system, edit owned products, and manage product information.

- A grid will show the following columns: Product, Description, Price.
- The "add product" button will open an empty add/edit product form for creating a new product.

- The "delete product" button will delete the currently selected row after a confirmation dialogue.
- The search box will filter the list by using the product name.
- The grid can be sorted by clicking on any column header.

4. Add/Edit Product Form: This is the form where the client is able to create new products or edit existing products.

- Include the following editable fields: Product Name, Description, Price.
- "Save" button for saving changes. These changes must be reflected in the table and the database.
- All fields must be validated when attempting to save. No field should be empty, except for the description which is optional. Price must be a valid amount in PHP.



5. Order Management Tab: This is the tab where vendors can view and manage orders for their products.

- A grid will show the following columns: Order ID, Product, Quantity, Unit Price, Total Price, Order Date, Status.
- The search box will filter the list by using either the Order ID or Product.
- The user can also filter the list with two date pickers (from & to), and a status combo box (Pending, Completed, Canceled).
- The "Complete Order" and "Cancel Order" button will mark the selected order as completed or canceled after a confirmation dialogue box. This button is only enabled for pending orders.
- The grid can be sorted by clicking on any column header.